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Bogies-VF.CONCEPT

SFRS for Zefiro-Europe

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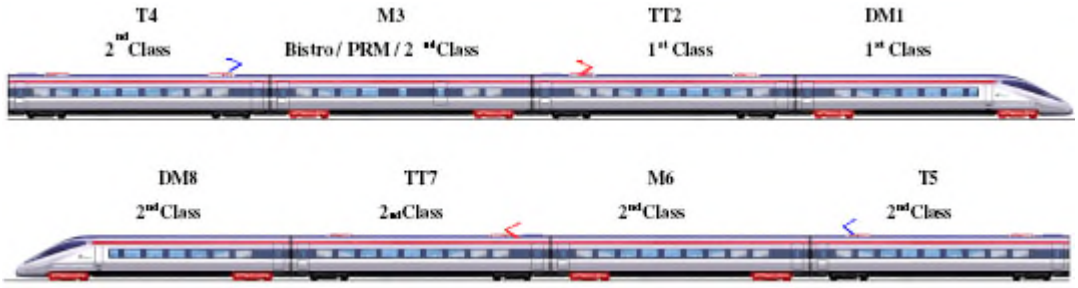
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1 Introduction

This document is a function specification for the control of the *XXX* system for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
TCMS	Train Control and Management System
IMS	Instability Monitoring System
ALS	Active Lateral Suspension
CBM	Condition Based Maintenance
WFL	Wheel flange lubrication
HABD	hot axlebox detection
SPM	Shock-Pulse Method
HABD	Hot axle box detection
DNRA	Detection of non-rotating axle
CAN	Controller Area Network
TCMS	Train Control Management System
TRS	Technical requirements specification
SFRS	System Function Requirement Specification
ICD	Interface Control Document
IP	Internet Protocol
GPS	Global Positioning System
FVDS	Functional Vehicle Design Specification

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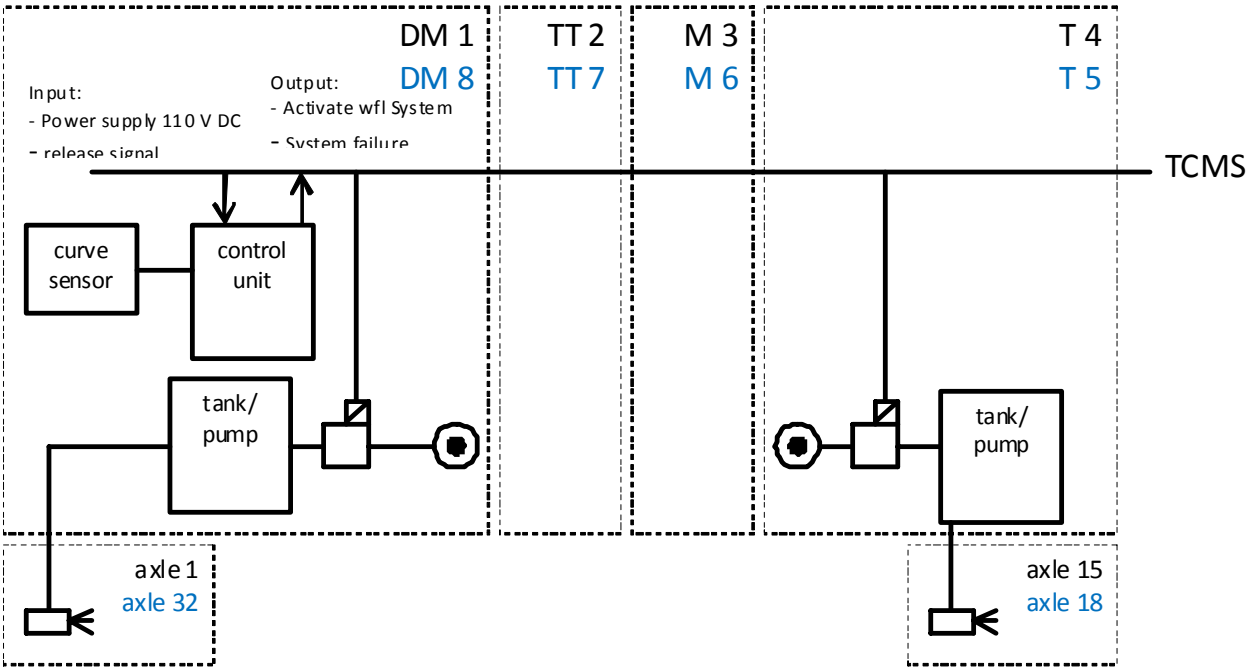
2 System Overview

This SFRS covers the flange lubrication, hotbox detection, instability monitoring system, active lateral suspension and diagnostics & condition based maintenance. A separate overview is given for every system.
The Bogie Controller manages the ALS function of both bogies of the same car to improve the comfort levels.
The CBM Controller monitors the conditions of certain bogie components to optimise the maintenance interventions by detecting degradations.

2.1 System Configuration

2.1.1 Flange Lubrication

WFL schematic



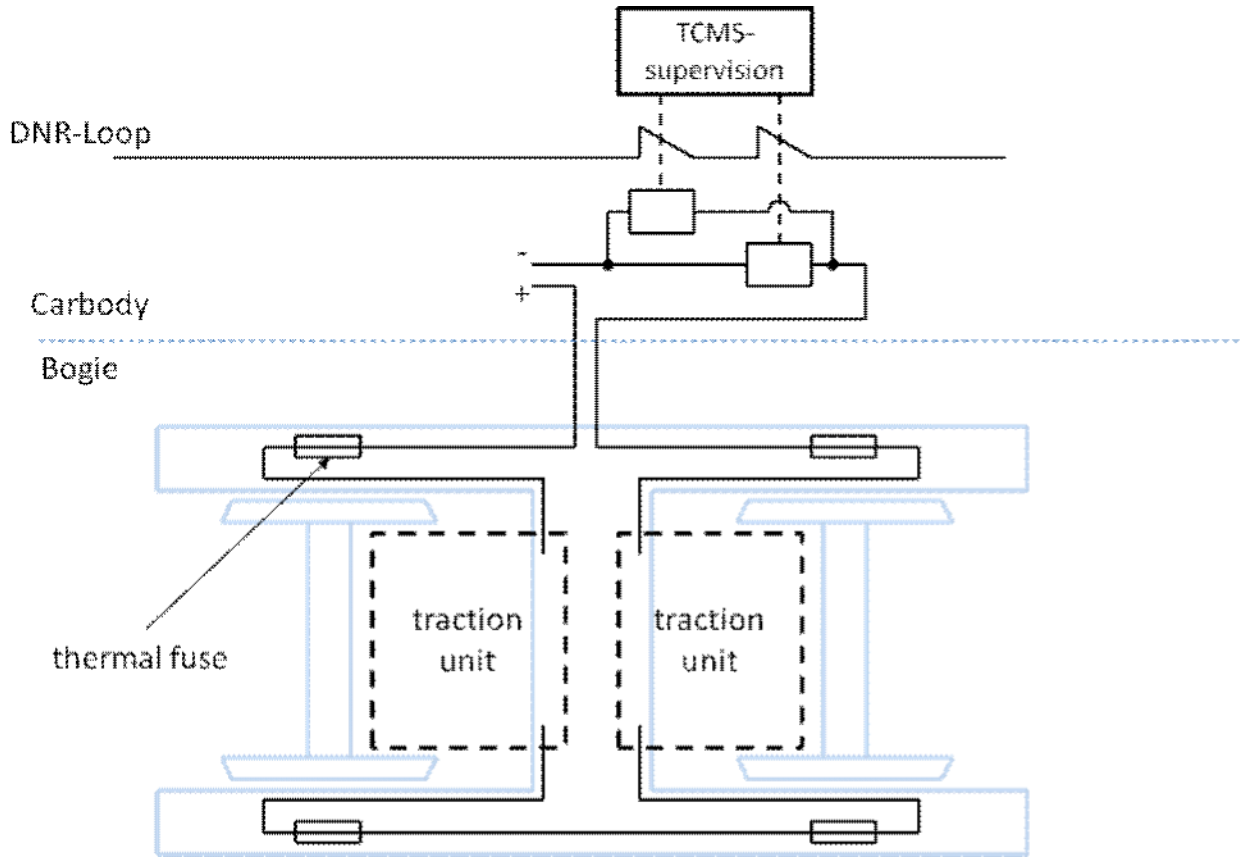
The wheel flange lubrication system consists of two separate and independent units, each for one running direction. Each unit consists of one control unit including curve sensor and two lubrication units. One for axle 1 and one for axle 15 and accordingly axle 32 and axle 18.

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2.1.2 Hotbox Detection

Hot axlebox detection schematic - Safety loop:



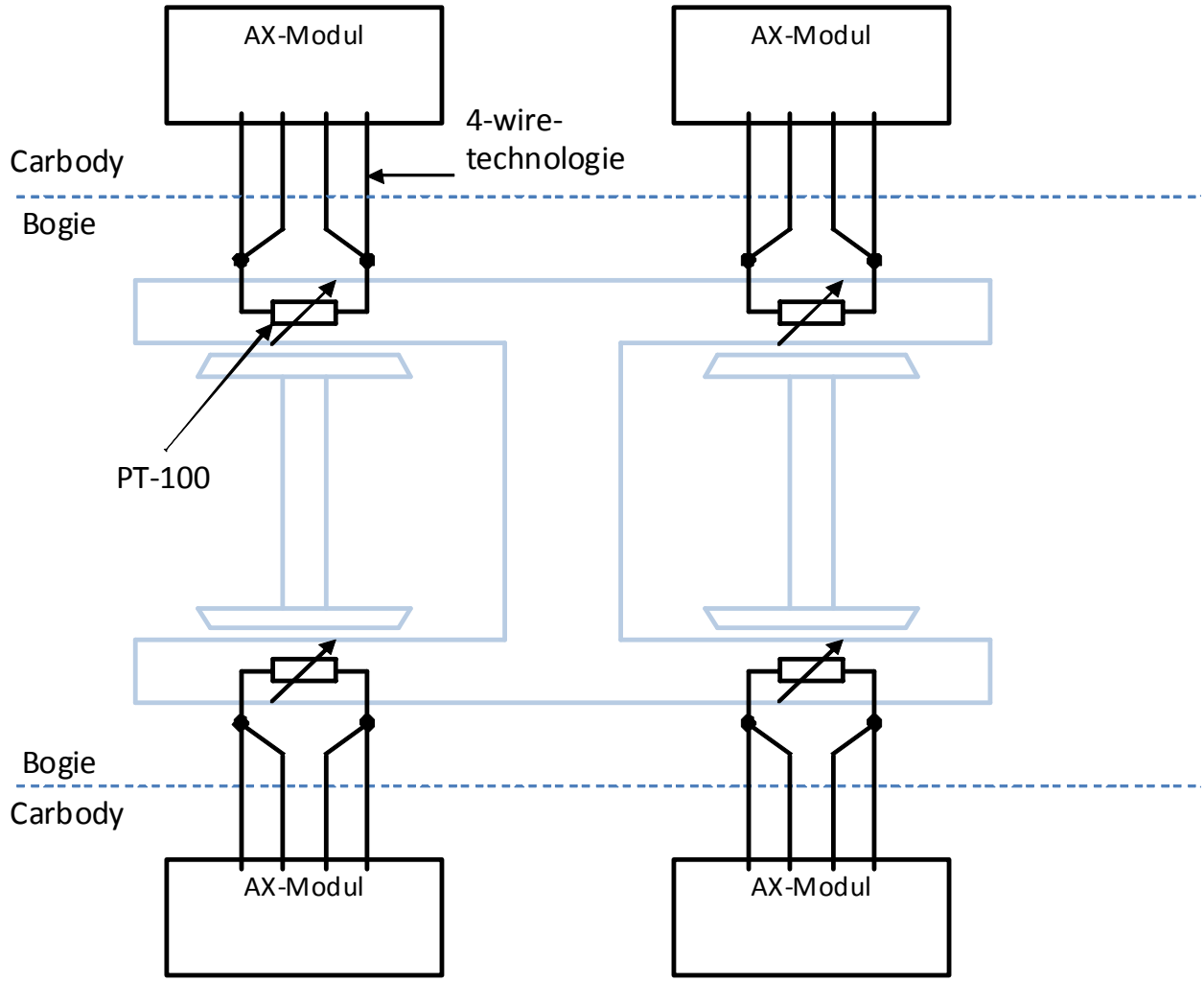
Hot axle box detection safety loop

Remark: The relais which open the DNRA loop are connected to the HABD safety loop of one specific bogie.

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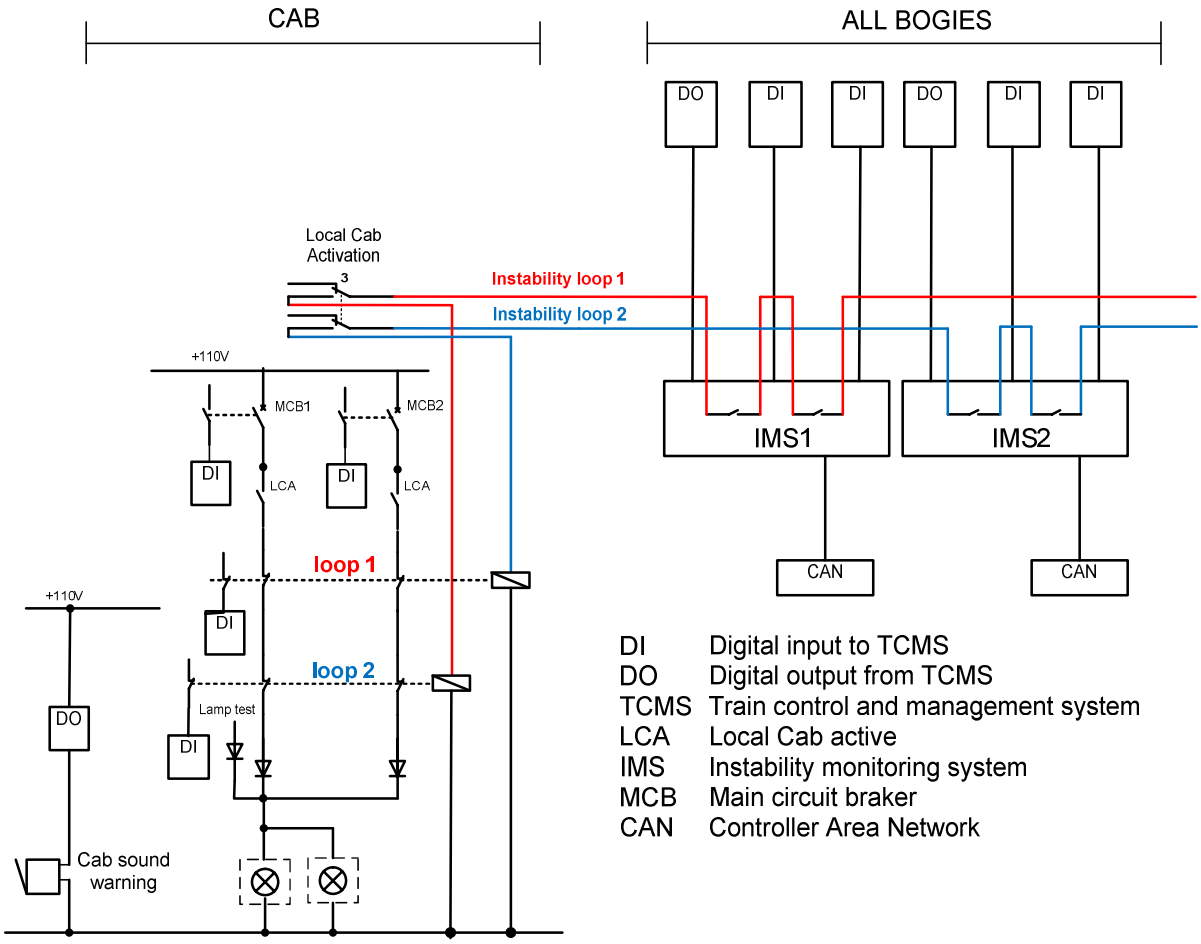
Hot axlebox detection schematic - Monitoring loop:



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2.1.3 Instability Monitoring

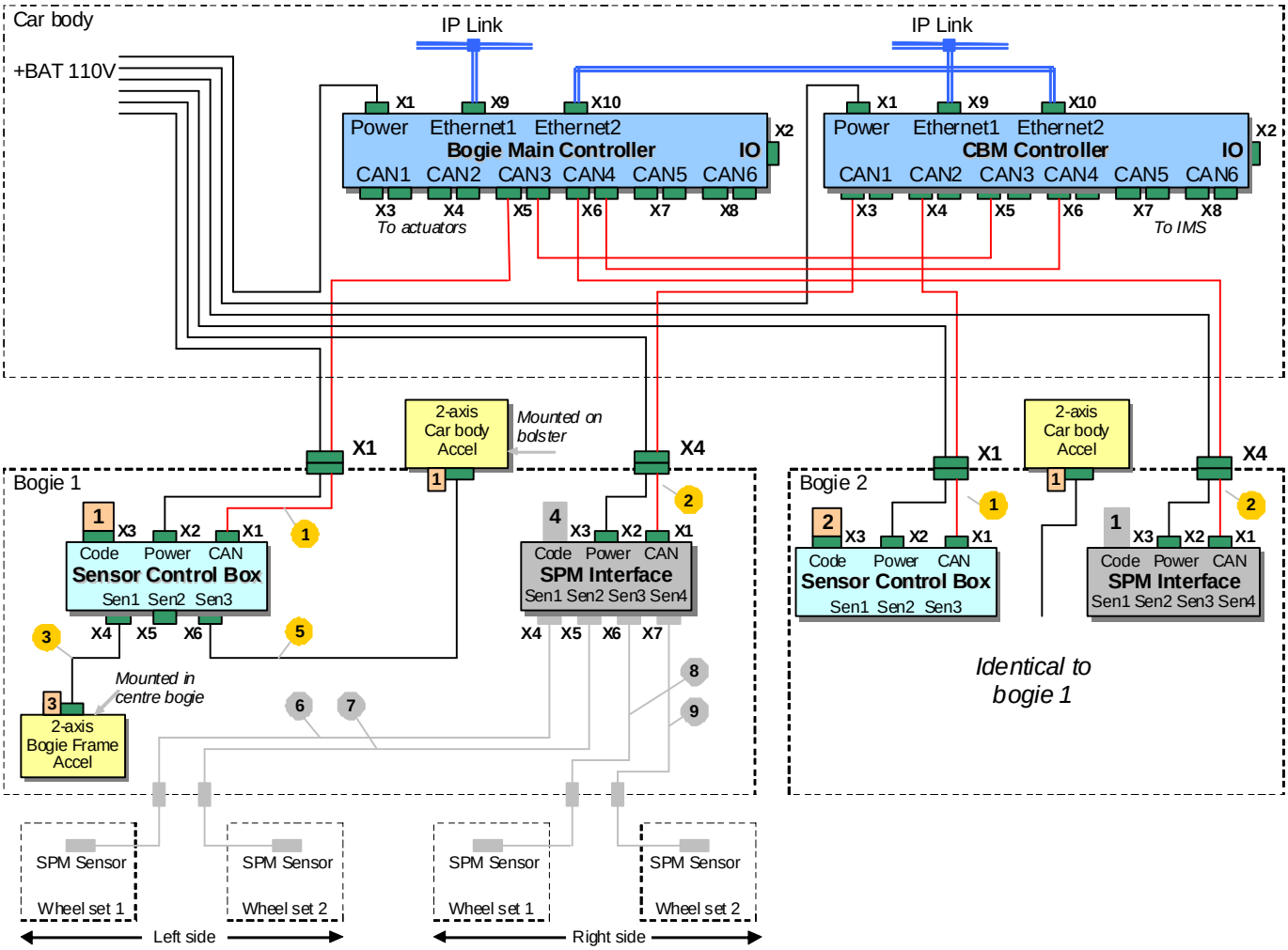


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2.1.4 Active Lateral Suspension

Sensor configuration: ALS-CBM

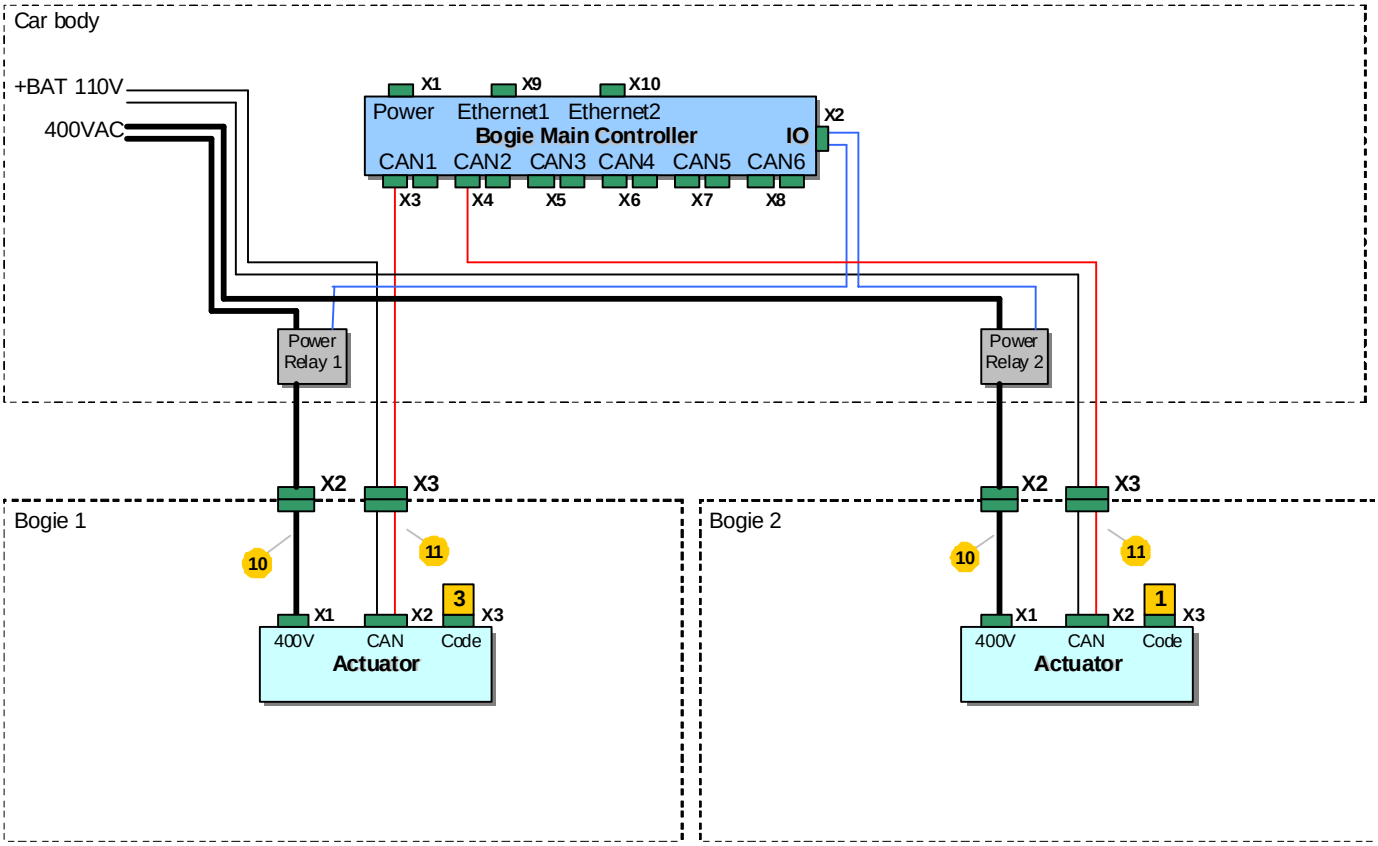


Note: the SPM interface and the SPM sensor are considered as an option

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Actuator circuit: ALS



2.1.5 Diagnostics & CBM

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3 Functional requirements

3.1 Flange Lubrication

3.1.1 Function Design

It is the purpose of the wheel flange lubrication system to reduce the wear between wheel flange and rail by using specialized lubricants. The lubricant is applied by means of air as a fine spray onto the wheel flange via single medium spray nozzles. As described in 2.1.1. the WFL consists of two symmetric units. Only one unit is described in this part. All requirements are accordingly valid for the second unit. The amount of lubricant per spray cycle that is applied on the wheel flange and activation of a spray cycle is controlled by the WFL control unit.

3.1.1.1 Normal condition

If all requirements on the train side are fulfilled, the flange lubrication is active. An active WFL system releases a spray signal if the integrated curve sensor. This signal is linked to the solenoid valves via TCMS. They operate when the signal is applied and a spray of grease is released.

3.1.1.2 Failure cases

1. Lubrication applied when not ordered
2. Lubrication not applied when ordered
3. Failure of flange lubrication control unit

3.1.1.3 Failure detection

1. No detection of failure during service possible. Detection of failure during normal maintenance by monitoring the amount of lubricant that is refilled.
2. No detection of failure during service possible. Failure detected during maintenance by use of integrated test function.
3. Event of WFL failure shall be generated and stored in TCMS

3.1.1.4 Failure handling

1. System repaired when failure detected during scheduled maintenance.
2. System repaired when failure detected during scheduled maintenance.
3. System shall be repaired during next scheduled maintenance

3.1.1.5 Failure effect on operation

1. No effect
2. No effect
3. No effect

3.1.2 Requirements

ID	2F_03.Zefiro-Europe.TRS.14
Description	Wheel flange lubrication shall be installed on at least two axle of the leading bogie. It shall operate curve dependently.
Safety level	N/A

One WFL unit consists of one control unit(inc. curve sensor) located in the cab and two lubrication units located in the underframe in front of axle 1 and axle 15. The tanks should be installed where quick refill by means of a filler coupling and the use of a separate filler pump or after opening the filler lid is possible.
The length of the tubing system from the tank to the nozzles shall be as short as possible.
The control unit installed in the cab must be easily accessible

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Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.167	A 'spray impulse' signal shall be generated when a curve is detected by the WFL system.	Derived Requirement	4S_03.06.02.-.Flange Lubrication Systems		N/A
2F_03.Zefiro-Europe.CONCEPT.156	The 'spray impulse' signal shall be linked from the control unit to TCMS.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_03.Zefiro-Europe.CONCEPT.279	The solinoid valves on the lubrication units on axle 1 in the DM 1 car and axle 15 in the T4 car shall be linked to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_03.Zefiro-Europe.CONCEPT.280	If the 'spray impulse' signal is set by the control unit, both solinoid valves of the active wfl system shall be activated.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_03.Zefiro-Europe.CONCEPT.325	The control unit installed in the cab must be easily accessible	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A

ID	2F_03.Zefiro-Europe.TRS.11
Description	The lubrication shall be enabled if – speed v > 3 km/h – sanding not active – train speed v <= 160km/h
Safety level	N/A

The input requirement specifies the case when the wheel flange lubrication system must be enabled. This is derived to TCMS Software to generate an 'enable wheel flange lubrication' signal. Therefore only one signal has to be send to the lubrication control unit. Which of the control units shall receive the 'enable wheel flange lubrication' signal depends on the running direction. It always has to be the control unit in the leading cab of the train. The cab activation signal can't be used for this purpose because in this case the wheel flanges of a second train in multi traction would not be lubricated. That means, that the 'enable wheel flange lubrication' is only valid on train level.

Additionally to the enabling signal a fault signal must be added to generate an event in the diagnostic, when a detectable fault in the wheel flange lubrication unit occurs.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.108	The output of TCMS 'enable wheel flange lubrication' shall be set for the control unit in the leading cab of the train when – speed v > 3 km/h – sanding not active – train speed v <= 160km/h	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_03.Zefiro-Europe.CONCEPT.350	The 'enable wheel flange lubrication' signal shall to be linked from TCMS to the Flange lubrication control box.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_03.Zefiro-Europe.CONCEPT.157	If the wheel flange lubrication control unit input 'enable wheel flange lubrication' is set the wheel flange lubrication system shall be enabled	Derived Requirement	4S_03.06.02.-.Flange Lubrication Systems		N/A
2F_03.Zefiro-Europe.CONCEPT.158	If the output of the wheel flange lubrication control unit 'wheel flange lubrication fault' is set to 0 for >2 seconds an event shall be set in the diagnostic system. If the output is set to 1 the system shall be considered as Ok.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_03.Zefiro-Europe.CONCEPT.351	The 'wheel flange lubrication fault' signal shall to be linked from the Flange lubrication control box to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_03.Zefiro-Europe.CONCEPT.159	The output of the wheel flange lubrication control unit 'wheel flange lubrication fault' shall be set to 0 when any kind of failure is detected by the control unit. In case of no failure the signal shall be set to 1.	Derived Requirement	4S_03.06.02.-.Flange Lubrication Systems		N/A
2F_03.Zefiro-Europe.CONCEPT.282	The description of an occured failure shall be displayed on the control unit	Derived Requirement	4S_03.06.02.-.Flange Lubrication Systems		N/A

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ID	2F_03.Zefiro-Europe.TRS.16
Description	Spraying of the grease shall be triggered by a pulse from the lateral accelerometer if a curve is detected. The wheel/rail contact area shall not be contaminated.
Safety level	N/A

The grease of the wheel flange lubrication system must be sprayed when the limit value of the curve sensor is reached for more than a specific period of time to avoid a spray to be triggered by other short-time exceedings of the limit value.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.111	The spraying of the grease shall be triggered when the curve signal of the curve sensor is set for >=2s(tbd)	Derived Requirement	4S_03.06.02.-.Flange Lubrication Systems		N/A
2F_03.Zefiro-Europe.CONCEPT.278	The detection level of the curve sensor shall be adaptable	Derived Requirement	4S_03.06.02.-.Flange Lubrication Systems		N/A
2F_03.Zefiro-Europe.CONCEPT.161	The spraying of the grease shall be triggered in a defineable periode	Derived Requirement	4S_03.06.02.-.Flange Lubrication Systems		N/A
2F_03.Zefiro-Europe.CONCEPT.162	When triggered the grease shall be sprayed for a definable periode of time	Derived Requirement	4S_03.06.02.-.Flange Lubrication Systems		N/A
2F_03.Zefiro-Europe.CONCEPT.163	The wheel/rail contact area shall not be contaminated	Derived Requirement	4S_03.06.02.-.Flange Lubrication Systems		N/A

ID	2F_03.Zefiro-Europe.TRS.20
Description	Test of the lubrication system shall be possible in maintenance mode by using: – a push button on the lubrication control unit
Safety level	N/A

A selftest function has to be integrated in the WFL system. It should be possible to release a spray of grease in standstill to visually check the amount of grease sprayed on the wheel flange

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.114	A button and function must be integrated in the lubrication control unit to release a spray of grease for test purpose.	Derived Requirement	4S_03.06.02.-.Flange Lubrication Systems		N/A

3.2 Hotbox Detection

3.2.1 Function Design

The hot axle box detection(HABD) monitors the temperature of each wheelset bearing which indicates damage of the bearings that may lead to critical accidents. Therefore this function is highly safety relevant.
The supervision is done by two separate onboard systems.

- Measurement of the temperature by a Pt 100 Sensor that is supervised by TCMS - temperature of all bearings supervised by TCMS. If certain temperature thresholds are exceeded, events shall be stored in the diagnostic system of the train.
- Detection of critical temperature threshold by thermal fuses which opens a safety loop of the effected bogie and leads to the activation of an alarm

3.2.1.1 Normal condition

Continuous measurement of the axle box bearing temperatures. If over-temperature is detected for a bogie bearing an event must be set according to the rate of the temperature.
If the temperature of a bearing is to high, a thermal fuse melts and opens the DNRA brake loop

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3.2.1.2 Failure cases

1. Overtemperature is present but not detected by Pt100
2. Overtemperature is not present but indicated by Pt100.
3. Thermal fuse melts or bogie loop is broken but no overtemperature is present
4. Thermal fuse melts, but a safety relays to open DNRA brake loop is welded closed.
5. Thermal fuse doesn't melt but Pt100 detects overtemperature.

3.2.1.3 Failure detection

1. Temperature supervision by thermal fuse. The check of temperature doesn't show which axle box is effected but the TCMS supervision of the bogie loop indicates which bogie is effected. So each temperature supervision of the effected bogie hast to be checked during repair.
2. Event will be set in the diagnostic system.
3. Train considers the situation as a real hot box which leads to the repair of the train. The failure is detected during the repair of the effected bogie.
4. TCMS supervises both relays and if they are not aligned a diagnostic event is set - Implausibility of safety relays
5. Event in the diagnostic system due to detection of Pt100.

3.2.1.4 Failure handling

1. Replacement of sensor during repair. Repair of axle box/bearing during unplanned workshop visit
2. Replacement of sensor and check of axle box during next scheduled maintenance.
3. Check/repair of all bogie loop parts during unplanned workshop visit.
4. Repair of axle box/bearing during unplanned workshop visit.
   Replacement/check of safety relays in the unplanned workshop visit.
5. Axle box/bearing checked/repaired during next scheduled maintenance.

3.2.1.5 Failure effects on operation

1. Train is stopped due to normal function of thermal fuse.
2. No effect
3. Train is stopped due to normal function of safety loop(opening of DNRA-loop)
4. Train is stopped due to normal function of safety loop(opening of DNRA-loop)
5. No effect

3.2.2 Requirements

3.2.2.1 Hot axle box detection safety loop

ID	2F_03.Zefiro-Europe.TRS.67
Description	The temperature of all axle bearings shall be supervised by hot box detectors installed on board the train.
Safety level	SIL 2

ID	2F_03.Zefiro-Europe.TRS.56
Description	If any hotbox detector reports "over temperature" then the driver shall be notified.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.255	All thermal fuses of a bogie must be connected in serial to a car safety loop which is deenergized when one thermal fuse melts.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_03.Zefiro-Europe.CONCEPT.349	A thermal fuse shall be installed in every axle box.	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 2

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2F_03.Zefiro-Europe.CONCEPT.256	The car loop shall be connected in parallel to two power saftey relais which open in series the DNRA safety line(to be verified with brake control system).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_03.Zefiro-Europe.CONCEPT.367	Activating a sealed switch, it shall be possible to bypass HABD car loop (separately in each coach) avoiding DNRA trainline to be opened by car loop relays.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_03.Zefiro-Europe.CONCEPT.368	In case HABD bypass car switch is activated, an event shall be set (warning to the driver)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_03.Zefiro-Europe.CONCEPT.392	If any HABD bypass car switch is reported to be activated, speed limitation shall be set to the driver reporting 200km/h. In case, the car is in "DNRA - BMS Local Monitoring" (as defined in 2F_05.02 concept 4117) status the speed limit shall be removed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_03.Zefiro-Europe.CONCEPT.318	Both safety relais shall be supervised by TCMS.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_03.Zefiro-Europe.CONCEPT.257	If one of both safety relais is open, an event shall be set stating which car is effected. Additionaly the current temperatures from the Pt100 supervisions of every thermal fuse sensor shall be stored. That includes 8 signals from axle box sensors per car and 8 signals from gearbox sensors per motor car.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

3.2.2.2 Temperature supervision

ID	2F_03.Zefiro-Europe.TRS.102
Description	If any hotbox detector reports "over temperature" then an event shall be set in the diagnostic system.
Safety level	N/A

ID	2F_03.Zefiro-Europe.TRS.109
Description	If any gearbox sensor detects "over temperature" then an event shall be set in the diagnostic system and the driver shall be notified
Safety level	

The temperature detection shall have two thresholds to distinguish between two different damage states of the bearings according to EN 15437-2.
The temperature signal provided by Propulsion System is filtered and smoothed with respect to the original analog signal

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.60	Pt100 Sensor shall be installed in every wheel axle box.	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 0
2F_03.Zefiro-Europe.CONCEPT.254	Temperature signal and axle box identification shall be displayed on IDU. All gearbox temperatures shall displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

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2F_03.Zefiro-Europe.CONCEPT.388	Gearboxes temperature signals (Motor car only) shall be supervised by TCMS. Following threshold shall be implemented, monitoring the highest valid value for each gearbox: 1) Temp >= 135 °C for more than 30 s after a low pass filter with T=8sec: an event shall be set and a warning shall be displayed on IDU 2) Temp >= 175 °C for more than 30 s after a low pass filter with T=8sec: an event shall be set and an alarm shall be displayed on IDU Each event shall report the information about which bogie and which gearbox is affected	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_03.Zefiro-Europe.CONCEPT.364	TCMS shall monitor single shaft of each gearbox and set following events based on the difference between the two temperature sensors. The delta temperature for each shaft will be calculated only if the temperatures of both bearing of the shaft are > 40°C. **Motor side bearings (fast side)** – Maintenance message: Difference between both sensors |T1-T2|>30°C for more than 30 s, after a low pass filter with T=8sec, maintenance message will be issued to the maintainer. – Alarm: Difference between both sensors |T1-T2|>40°C for more than 30 s, after a low pass filter with T=8sec, the Alarm will be issued on driver's IDU only if the warmest bearing of the shaft has temperature >100°C Alarm will require to the driver to stop immediately **Wheelset side bearing (slow side)** – Maintenance message: Difference between both sensors |T1-T2|>25°C for more than 30 s,after a low pass filter with T=8sec, maintenance message will be issued to the maintainer. – Alarm: Difference between both sensors |T1-T2|>35°C for more than 30 s, after a low pass filter with T=8sec, the Alarm will be issued on driver's IDU only if the warmest bearing of the shaft has temperature >100°C alarm will require to the driver to stop immediately	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_03.Zefiro-Europe.CONCEPT.365	Every time any pre-alarm or alarm is set an event shall be shown on IDU according 2F_03.Zefiro_Europe.CONCEPT.364	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_03.Zefiro-Europe.CONCEPT.377	TCMS shall monitor single bearing of each gearbox and set following events based on the gradient evaluated over 30 sec. – Warning message: gradient > 7,5°C where the starting temperature is >80°C – Alarm: gradient > 12,5°C where the starting temperature is >80°C	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

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2F_03.Zefiro-Europe.CONCEPT.389	Every time any pre-alarm or alarm (both for axleboxes and garboxes) is set the related pictogram shall be shown on IDU. If an alarm follows a warning activation of the same criteria on the same sensor the transition between the pre-alarm and alarm pictogram has to be coninuous.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_03.Zefiro-Europe.CONCEPT.385	Every time any pre-alarm or alarm is set an event shall be shown on IDU according to 2F_03.Zefiro_Europe.CONCEPT.389.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_03.Zefiro-Europe.CONCEPT.380	Any alarm raised on the PT100 readings of the gearbox shall be latched (and so memorized also if the setting conditions are cleared). The memorization will be cancelled only by pressing a dedicated softkey on the related bogie page (available only at train standstill).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_03.Zefiro-Europe.CONCEPT.381	The use of the alarm reset softkey shall be recorded as an event.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_03.Zefiro-Europe.CONCEPT.378	If any gearbox alarm (based on PT100 readings) is triggered the TCMS shall activate a permanent traction block as long as the train speed >= 80 km/h. If the train speed is < 80 km/h the traction block can be restored (after positioning the traction lever in coasting).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_03.Zefiro-Europe.CONCEPT.383	In case the driver has reset the alarm the speed limitation shall not be active anymore even if the alarm setting conditions are still valid.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_03.Zefiro-Europe.CONCEPT.384	In case of activation of a new alarm (from another PT100 sensor) OR in case of re-activation of the the cleared alarm the speed limit shall be again active.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_03.Zefiro-Europe.CONCEPT.386	Any pre-alarm and/or alarm shall be shown on the wayside interface together with its "acknowledged/not acknowledged" information (additional checkbox).	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		SIL 0
2F_03.Zefiro-Europe.CONCEPT.359	If – MIO is on AND MIO declares a nth PT100 as valid. – independently from train speed the temperature of all **gearboxes** shall be evaluated: If the temperature is out of range (<-40°C or >400°C) a warning of sensor failure shall be stored in the diagnostic system including the identification of the effected gearbox bearing PT100. The above condition last until PT100 temperature are back into the plausibility range.	Derived Requirement	4S_09.03.05.-.TCMS Software		

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2F_03.Zefiro-Europe.CONCEPT.371	**PT100 gradient assessment to declare plausibility of gearbox bearing temperature.** 1) Ratio of gradient of PT100 bearing temperature when negative: – IF a cab is activated – IF the temperature gradient of nth bearing of mth gearbox is <0 – AND if the ratio between the module of the temperature gradient of the nth bearing and the biggest module among all the remaining homologous bearings (having temperature gradient < 0) of the trainset is >=4, – AND this condition shall be repeated for 3 times consecutively the PT100 of the nth bearing has to be declared as faulty and their temperature readings will be discarded. A PT100 declared as faulty will stay in this condition until next train reset. 2) Gradient of PT100 bearing temperature when positive: – IF a cab is activated – IF the temperature gradient of nth bearing of mth gearbox is >0 – AND IF the temperature gradient of the nth bearing of mth gearbox is > 10 °C/s, the PT100 of the nth bearing has to be declared as faulty and their temperature readings will be discarded. A PT100 declared as faulty will stay in this condition until next train reset. 3) Gradient of PT100 bearing temperature when negative: – IF a cab is activated – IF the temperature gradient of nth bearing of mth gearbox is <0 – AND the temperature gradient of the nth bearing of mth gearbox is < -10 C°/s, the PT100 of the nth bearing has to be declared as faulty and their temperature readings will be discarded. A PT100 declared as faulty will stay in this condition until next train reset. The above cases shall generate a maintenance event to report the identification of the affected gearbox bearing PT100 Note: the temperature gradient for the bullet 1 shall be calculated with dt of 10s	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

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2F_03.Zefiro-Europe.CONCEPT.376	**PT100 gradient assessment to declare plausibility of axlebox bearing temperature.** 1) Ratio of gradient of PT100 bearing temperature when negative: – IF a cab is activated – IF the temperature gradient of nth bearing of mth axlebox is <0 – AND if the ratio between the module of the temperature gradient of the nth bearing and the biggest module among all the remaining homologous bearings (having temperature gradient < 0) of the trainset is >=4, – AND this condition shall be repeated for 3 times consecutively the PT100 of the nth bearing has to be declared as faulty and their temperature readings will be discarded. A PT100 declared as faulty will stay in this condition until next train reset. 2) Gradient of PT100 bearing temperature when positive: – IF a cab is activated – IF the temperature gradient of nth bearing of mth axlebox is >0 – AND IF the temperature gradient of the nth bearing of mth axlebox is > 10 °C/s, the PT100 of the nth bearing has to be declared as faulty and their temperature readings will be discarded. A PT100 declared as faulty will stay in this condition until next train reset. 3) Gradient of PT100 bearing temperature when negative: – IF a cab is activated – IF the temperature gradient of nth bearing of mth axlebox is <0 – AND the temperature gradient of the nth bearing of mth axlebox is < -10 C°/s, the PT100 of the nth bearing has to be declared as faulty and their temperature readings will be discarded. A PT100 declared as faulty will stay in this condition until next train reset. The above cases shall generate a maintenance event to report the identification of the affected axlebox bearing PT100 Note: the temperature gradient for the bullet 1 shall be calculated with dt of 10s	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_03.Zefiro-Europe.CONCEPT.271	The Pt100 sensor shall be connected to an AX module, 4 wire technology must be used to measure the temperature.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_03.Zefiro-Europe.CONCEPT.117	If the axle box temperature signal exceeds +95 °C for more than 30 s, an event shall be set in the diagnostic system and displayed on the IDU including the identification of the effected axle box.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

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2F_03.Zefiro-Europe.CONCEPT.362	If the axle box temperature signal exceeds +110°C for more than 30 s, an alarm event shall be set in the diagnostic system and displayed on the IDU including the identification of the affected axle box.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_03.Zefiro-Europe.CONCEPT.240	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_03.Zefiro-Europe.CONCEPT.241	If – MIO is on AND MIO declares a nth PT100 as valid. – independently from train speed the temperature of all axle boxes shall be evaluated: If the temperature is out of range (<-40°C or >400°C) a warning of sensor failure shall be stored in the diagnostic system including the identification of the affected axlebox bearing PT100. The above condition last until PT100 temperature are back into the plausibility range.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

3.2.2.3 Test Function

ID	2F_03.Zefiro-Europe.TRS.108
Description	A test function shall be implemented for the hot box detection
Safety level	

To ensure the functionality of the hot axle box detection safety loop a test function shall be integrated.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.339	A relay shall be installed in each car and shall trigger a "break" contact of each bogie of this car.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		
2F_03.Zefiro-Europe.CONCEPT.343	The test relay shall be linked to TCMS digital output.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		
2F_03.Zefiro-Europe.CONCEPT.344	The Hot Box Detection Test/ Bearing monitoring test shall be triggered by the DNRA-BMS-Test(Brake Control SFRS Req. 2F_05.02.3520).	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_03.Zefiro-Europe.CONCEPT.315	A permanent supervision of the HABD car loop relays shall be implemented in TCMS If one supervision signal shows a different state of the relay than the other for more than 1 second, an EVENT shall be set in the diagnostic system stating implausibility of the hot box detection safety relays.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_03.Zefiro-Europe.CONCEPT.347	The result of the HABD-test shall be presented to the driver on the IDU. A failed test indication on the IDU shall be reset by a successful additional test.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	
2F_03.Zefiro-Europe.CONCEPT.370	The status of HAB loop relays shall be presented, per car, to the driver on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	

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3.3 Instability Monitoring

3.3.1 Function Design

The Instability monitoring system supervises the lateral acceleration of a bogie during the service of the V300ZEFIRO – ETR1000 high speed train.
The bogie state that shall be considered as instability is defined in the HS RST TSI and implemented in the presented system. The detection is based on the measurement of the lateral acceleration of the bogie frame above the centreline of the wheelset.
In case the defined conditions are present (10 consecutive lateral acceleration cycles of 0,8g and a frequency between 3 and 9 Hz) the bogie mounted supervision devices (IMS boxes) indicate instability.
For availability reasons two IMS boxes are mounted diagonally on each bogie. They are stand-alone systems which provide via digital output to TCMS and via safety relays that open one of the train lines the detected instability status.
According to HS RST TSI the driver is informed by a visual signal on the drivers desk in case of instability and in addition to that a buzzer is sounded inside the cab. The activation of the indication to the driver is monitored by TCMS

The function of the IMS is to indicate instability to the driver when the TSI criteria for instability is exceeded. As soon as instability is present the driver is informed via the IDU.
For diagnostic purposes the type of instability is divided into 3 levels.

- Level 0
- Level 1
- Level 2

The Level 0 warning indicates short term instability cases which are present not longer than 1 second. This cases in which the lateral acceleration amplitude drops below the TSI relevant value of 0,8g within 1 second lead to investigative maintenance actions for bogie parts and the detection equipment.
Level 1 and Level 2 identify longterm (more the 1 second) instability cases. If the magnitude of the instability is below 15 m/s² the incident is stored in TCMS as level 1 Instability. If the magnitude is higher than 15 m/s², the incident is stored as level 2.

3.3.1.1 Normal condition

Instability is detected by both IMS-units of a bogie and a instability warning or alarm is displayed on the driver's desk according to the specification of the warning levels.

3.3.1.2 Failure cases

1. Instability is present but not detected.
2. No instability is present, but falsely indicated by one box of the bogie?

3.3.1.3 Failure detection

1. Not possible to detect by the driver. In case of instability detection by the train crew(noise, vibration) train driver must be alerted.
2. It must be considered as an 'Instability detection', eventual IMS fault can be detected during IMS test

3.3.1.4 Failure handling

1. Effected bogie has to be checked during corrective maintenance
2. Train acts according to the level. Boxes have to be repaired during unplanned workshop visit.

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3.3.1.5 Failure effect on operation

1. No effect from train. Driver has to act according to operational manual(actions have to be aligned with customer)
2. Train is limited to 220 km/h

3.3.2 Requirements

This part describes the functions of the instability monitoring system unit that is located in the bogie and are related to all instability level warnings described below.

ID	2F_03.Zefiro-Europe.TRS.82
Description	An on board instability monitoring system shall be fitted. The detection of instability shall be based on acceleration measurements taken on the bogie frame. An instabilty warning or alarm shall advise the driver to reduce the speed. The criteria for activation of this alarm shall be as defined in <<HS RST TSI 2008>> clause 4.2.3.4.2 d2.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.65	An on board instability monitoring system with 2 redundant detection units per bogie shall be installed.The units shall be installed diagonally in the bogie.	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 2
2F_03.Zefiro-Europe.CONCEPT.169	An IMS unit shall include a lateral accelerometer that shall be placed on the bogie frame above the centreline of the wheelset.	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 2
2F_03.Zefiro-Europe.CONCEPT.170	A lateral acceleration of a>=8 m/s² at a frequency between 3 to 9 Hz and for more than 10 consecutive cycles shall be considered as instability by the IMS unit and 'Instability Detection' shall be set.	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 2

ID	2F_03.Zefiro-Europe.TRS.95
Description	Two digital inputs shall be provided to indicate 4 states to TCMS: – System OK – High Level Detection (level 2) – Instability Detection (level 0 and level 1) – System Error
Safety level	SIL 0

Apart from the link to the trainline a IMS unit shall be connected to TCMS via 2 digital signals which provide 4 states to TCMS.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.129	2 Digital inputs per IMS unit shall provide the IMS status to TCMS: – System OK (1 1) – High Level Instability Detection (level 2) (1 0) – Instability Detection (level 0 and level 1) (0 1) – System Error (0 0)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_03.Zefiro-Europe.CONCEPT.234	The three levels of IMS indication shall be sent to TCMS via the I/O link: – System OK – High Level Instability Detection (level 2) – Instability Detection (level 0 and level 1) – System error	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 0
2F_03.Zefiro-Europe.CONCEPT.179	System Error shall be set when an internal error of any kind occured in the IMS unit	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 0
2F_03.Zefiro-Europe.CONCEPT.181	High Level Detection shall be set when instability Detection is active and a lateral acceleration peak of a>=15 m/s² occured.	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 0

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2F_03.Zefiro-Europe.CONCEPT.182	System OK shall be set if non of the following – System Error or – High Level Detection or – Instability Detection are set.	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 0
2F_03.Zefiro-Europe.CONCEPT.177	2 digital inputs and 1 output shall be linked from every IMS unit (per bogie) to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

ID	2F_03.Zefiro-Europe.TRS.91
Description	For IMS, the detection threshold levels and number of consecutive cycles shall be adaptable.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.120	The detection treshold levels (lateral acceleration high detection level and low detection level) and the number of consecutive cycles of the IMS system shall be adaptable.	Derived Requirement	4S_03.-.Bogies and Running Gear		

ID	2F_03.Zefiro-Europe.TRS.93
Description	For availability reasons there shall be two redundant train-lines and two IMS units per bogie. The two IMS units per bogie shall be connected to different train lines.
Safety level	N/A

ID	2F_03.Zefiro-Europe.TRS.92
Description	The safety relays of all IMS units of the train shall be connected in serial connection to respective train-lines to communicate the instability warning to a lamp (and/or buzzer) in the cab. The trainline shall be of fail safe type i.e. the trainline is de-energised by the safety relays breaking the loop.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.123	2 trainlines shall be integrated in train. They shall connect the IMS boxes according to the schematic in 2.1.3. to the indication lamps on the drivers desk	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

The two trainlines shall be called "loop 1" and "loop 2" according to the schematic in 2.1.3.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.233	Both IMS loops shall be supervised by TCMS. If a loop is open, a supervision signal shall be set.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_03.Zefiro-Europe.CONCEPT.215	If the "loop 2" supervision signal is set and no IMS box shows instability, an event shall be set in the diagnostic system.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_03.Zefiro-Europe.CONCEPT.331	If the "loop 1" supervision signal is set and no IMS box shows instability, an event shall be set in the diagnostic system.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_03.Zefiro-Europe.CONCEPT.321	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

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2F_03.Zefiro-Europe.CONCEPT.322	If both loops are open or If any IMS box shows "high level detection" or "instabiliy detection" or If both IMS boxes on a single bogie shows "System error" it shall be considered as instability by TCMS and the 'Instability present' signal must be set	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_03.Zefiro-Europe.CONCEPT.373	In case the status of IMS controller indicates High level instability Detection (1 0) OR Instability detection (0 1) AND the related IMS loop is energized for more than 30 s,the IMS box shall be excluded by TCMS (considered in "System Error") and the status shall be sent to IDU NOTE: the affected IMS box will stay in this condition until next train reset	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_03.Zefiro-Europe.CONCEPT.231	Two indication lamps shall be fitted in the driver's desk according to the schematic 2.1.3.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	SIL 2
2F_03.Zefiro-Europe.CONCEPT.175	The trainline shall be energised when no instability is detected.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_03.Zefiro-Europe.CONCEPT.176	If both trainlines are de-energised the lamp in the activated cab shall be active.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_03.Zefiro-Europe.CONCEPT.174	If instability is detected by an IMS box, the trainline must be broken by a safety relay in the IMS box when the additional requirements of level 1 or level 2 instability are fullfilled.	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 2
2F_03.Zefiro-Europe.CONCEPT.353	If any IMS box indicates 'high level indication' or 'instability indication' a diagnostic event shall be set (indicating the affected box and its status)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_03.Zefiro-Europe.CONCEPT.374	If any IMS box indicates "system error" an warning event shall be shown to the Driver (indicating the affected box and its status)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	

3.3.3 Level 0 Warning

3.3.3.1 Function Design

As irregularities of the track may lead to short lateral oscillations of the bogie a level 0 warning was introduced into the IMS detection system. This state is only valid for 1 second.
When instability was detected by any IMS box, it is indicated to the train via the digital outputs of the boxes. The signal is read by TCMS and activates the instability alarm on the IDU.
After 1 second the safety relays in the IMS boxes open the train lines and the indication is escalated to a “Level 1” for which the actions are defined in section 3.3.4.
The following actions are performed during Level 0 warning:

- If instability is detected by an IMS box, the “Low level Detection” or “High level Detection” signal is set.
- If the status is still present after 1s, the safety relays of the affected IMS box are opened and the connected IMS train-line is deenergised.
- If any IMS box indicates “Low level Detection” or “High Level Detection” TCMS activates the instability alarm on the IDU and stores the Level 0 warning in the diagnostic system of the train including time and identification of the affected bogie.

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3.3.3.2 Requirements

ID	2F_03.Zefiro-Europe.TRS.68
Description	Instability warning level 0 shall be reported via diagnostics (TCMS) if a lateral acceleration of low level is detected (Fe => 8m/s² is detected between 3Hz and 9Hz for 10 consecutive cycles). The driver shall have the possibilty to decide upon actions considering the track status.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.212	If instability is detected a timer shall be started in the IMS unit	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 0
2F_03.Zefiro-Europe.CONCEPT.178	If – if the instability present signal is set (as described in 2F_03:Zefiro-Europe.CONCEPT.322) and – and the trainspeed is >220 km/h a instability alarm shall be displayed on the IDU for the driver including the identification of the affected car and the buzzer of the activated cab shall be sound. The event shall be stored in the diagnostic system of the train including time and identification. If trains speed is above 220km/h the traction shall be blocked as long as the instability is present (as described in 2F_03:Zefiro-Europe.CONCEPT.178).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_03.Zefiro-Europe.CONCEPT.183	If the instability present signal is set and train speed is below 220km/h an event shall be set in the diagnostic system	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

3.3.4 Level 1 Alarm

3.3.4.1 Function Design

To improve the reliability of the system two train-lines are implemented in the train. Each train-line connects in serie 16 IMS boxes (one out of two per bogie) to a relay in the cab. If instability is present on a bogie and is detected for more than 1 second on both IMS boxes of the bogie (see Level 0 warning in section 3.3.3), both train-lines are opened and lead to the activation of the indication lamp on the drivers desk and the activation of the main buzzer inside the cab.
This logic implements that the faulty indication of instability by one out of two IMS boxes per bogie does not lead to the activation of the indication devices on the drivers desk.
Only in case of a double failure (failure of one IMS box on both train-lines even if not on the same bogie) instability is falsely indicated to the driver when no instability is present.
The following actions are performed during Level 1 Alarm:

- The detection of instability for more than 1 second on one IMS box per train-line leads to the opening of both IMS train-lines
- The opening of both train-lines lead to the strictly wired activation of the indication lamp (see 2.1.3.).
- If both train-lines are opened TCMS sets the ‘instability present’ signal on the base of the digital inputs for **loop1** and **loop2** (see Figure 2.1.3.)
- When the “instability present” signal is set all bogies having at lease one “Low Level Detection” active are considered as affected by Level 1 instability.
- The identification of the affected bogie(s) is displayed on the IDU.
- The status “Level 1”, the identification of the affected bogie(s) and the time is stored in the diagnostic system.

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- The activation of the “instability present” signal is stored within TCMS.

3.3.4.2 Requirements

ID	2F_03.Zefiro-Europe.TRS.69
Description	Instability warning level 1 shall be indicated if the timer for an active level 0 warning exceeds 1s. This warning - now escalating into an alarm - shall be reported via the IMS trainlines.
Safety level	SIL 2

ID	2F_03.Zefiro-Europe.TRS.103
Description	Instability warning level 1 shall be indicated if the timer for an active level 0 warning exceeds 1s. This warning - now escalating into an alarm - shall be reported via the diagnostic system (TCMS).
Safety level	N/A

A safety relay shall be integrated in the IMS unit.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.213	If – instability is detected and – the IMS unit timer exceeds 1s the IMS unit safty relay shall break the IMS trainline.	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 2

If

– if the instability present signal is set (as described in 2F_03:Zefiro-Europe.CONCEPT.322)

and

– both IMS trainlines are open

and

– and the train speed is >220 km/h

an instability alarm shall be displayed on the IDU for the driver including the identification of the affected car and the buzzer of the activated cab shall be sound.
The event shall be stored in the diagnostic system of the train including time and identification.
If trains speed is above 220km/h the traction shall be blocked as long as the instability is present(as described in 2F_03:Zefiro-Europe.CONCEPT.178).

3.3.5 Level 2 Alarm

3.3.5.1 Function Design

Indication of instability “Level 2 Alarm”
If the lateral acceleration detected by an IMS box rises to 15m/s² the “High Level Detection” is set by the IMS box. The functionality for the activation of the indication devices is not affected by the Level of the Alarm.
The following actions are performed during Level 2 Alarm:

- When the “instability present” signal is set all bogies having at lease one “High Level Detection” active are considered as affected by Level 2 instability.
- The identification of the affected bogie(s) is displayed on the IDU.
- The status “Level 2 Alarm”, the identification of the affected bogie(s) and the time is stored in the diagnostic system.

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3.3.5.2 Requirements

ID	2F_03.Zefiro-Europe.TRS.77
Description	If in the mean time a peak acceleration is detected by IMS in excess of 15 m/s² (high level detected), then the alarm shall be reported via: – - the IMS trainline being deenergised
Safety level	SIL 2

ID	2F_03.Zefiro-Europe.TRS.104
Description	If in the mean time a peak acceleration is detected by IMS in excess of 15 m/s² (high level detected), then the alarm shall be reported via the IMS delivering the signal "high level" (level 2) to TCMS
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.80	if – instability is detected and – in the mean time a lateral acceleration of a>15m/s² is detected once the IMS unit safty relay shall break the trainline	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 2
2F_03.Zefiro-Europe.CONCEPT.275	if – instability is detected and – in the mean time a lateral acceleration of a>15m/s² is detected once the High Level Detection shall be set	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 0

If

– if the instability detection signal is set (as described in 2F_03:Zefiro-Europe.CONCEPT.322)

and

– both IMS trainlines are open

and

– one or both IMS boxes of a single bogie show "high level detection"

and

– the train speed is >220 km/h

a High level instability alarm event shall be set including the identification of the affected car and the buzzer of the activated cab shall be sound.
The event shall be stored in the diagnostic system of the train including time and identification.
If trains speed is above 220km/h the traction shall be blocked as long as the instability is present (as described in 2F_03:Zefiro-Europe.CONCEPT.178)

ID	2F_03.Zefiro-Europe.TRS.76
Description	If the instability level 2 alarm is triggered, train speed shall be limited. If speed is higher than the speed limit when the alarm is triggered,
Safety level	SIL 0

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3.3.6 Test

3.3.6.1 Function Design

3.3.6.2 Requirements

ID	2F_03.Zefiro-Europe.TRS.44
Description	The IMS shall perform an automatic self-test after Power-up
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.95	After power-up a selftest shall be started by every IMS unit. Including test of – sensors – detection algoritm – safety relays	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 2
2F_03.Zefiro-Europe.CONCEPT.221	when a failure is detected by the self test - system error - shall be set by the effected unit and the safety relay shall break the trainline.	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 2
2F_03.Zefiro-Europe.CONCEPT.223	if system error is set for TCMS must log the fault. This requirement is independent from the way the test is triggered.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_03.Zefiro-Europe.TRS.48
Description	A IMS self test shall be triggered by TCMS if a cab is activated and the train is standstill.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.141	When an end cab is activated and the train is at standstill, the selftest of the IMS units shall be activated by setting the test DO for both IMS boxes on a bogie at same time and checking the status of the related IMS loop. Each Car shall be checked one by one. TCMS shall trigger the test if at least one BC2 contactor is detected closed in each car.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_03.Zefiro-Europe.CONCEPT.235	If the signal to trigger a IMS self test is received from TCMS, the IMS shall perform its self test.	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 0

ID	2F_03.Zefiro-Europe.TRS.52
Description	The IMS train line during shall be open during test of the system.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.144	During a IMS selftest the trainline shall be opened.	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 2

ID	2F_03.Zefiro-Europe.TRS.94
Description	Every part shall be tested simultaneously on all IMS systems of the train. If the test is not successful, the driver is informed via the IMS lamp associated with the loop.
Safety level	SIL 2

ID	2F_03.Zefiro-Europe.TRS.101
Description	If the IMS test is not successful, the driver shall be informed via the diagnostic system (TCMS).
Safety level	N/A

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Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.147	When a test is triggered, every part shall be tested simultaneously on all IMS systems of the train. If the test is not succesful, the driver shall be informed via the IMS lamp associated with the loop.	Derived Requirement	4S_03.-.Bogies and Running Gear		SIL 2
2F_03.Zefiro-Europe.CONCEPT.237	The status of the IMS test shall be reported to TCMS using digital inputs of the TCMS	Derived Requirement	4S_03.-.Bogies and Running Gear		N/A
2F_03.Zefiro-Europe.CONCEPT.236	If the IMS test has not been successful (One IMS box or more states system error, IMS loop 1 does not open, IMS loop 2 does not open), an event shall be triggered in the diagnostic system.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.4 Active Lateral Suspension (ALS)

3.4.1 Function Design

The objective of the ALS function is to improve the comfort level of the train by active supervising the carbody position compared to the bogie and stabilize a centre position. It is managed by the bogie controller, a stand alone system which is connected to the TCMS to transmit diagnostic, configuration and status info and to receive the speed level, 400V present and the train configuration. The bogie controller manages the ALS system on both bogies of the same car. Additional to the active lateral suspension, each bogie is equipped with lateral buffers to assure running stability and safety in case of a failure of the ALS. That means, that ALS is not safety relevant.

3.4.1.1 Normal condition

The ALS has basically two tasks:

1. To create a lateral force in curves corresponding to lateral quasi-static acceleration times the pivot load of the bogie.
2. To reduce the vibration level in the carbody by creating a force counteracting the vibration transfer from bogie to carbody

3.4.1.2 Failure cases

1. Electrical power of the actuator lost - the actuator operates as a passive lateral damper
2. Zero force reference - the actuator will minimize force
3. Maximum force to one side - actuator will force bumbstop contact

3.4.1.3 Failure detection

1. tbd. - condition not safty relevant according to [6]
2. tbd. - condition not safty relevant according to [6]
3. tbd. - condition not safty relevant according to [6]

3.4.1.4 Failure handling

1. Repair during next scheduled maintenance
2. Repair during next scheduled maintenance
3. Repair during next scheduled maintenance

3.4.2 Requirements

ID	2F_03.Zefiro-Europe.TRS.88
Description	An ALS shall be provided on all bogies to ensure a good running behaviour and comfort.
Safety level	SIL 3

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ID	2F_03.Zefiro-Europe.TRS.96
Description	The ALS is integrated in the bogie controller which shall be connected to TCMS Ethernet to receive train data (e.g. speed) and to provide diagnostics and status reporting.
Safety level	SIL 3

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.195	The following status and configuration signals are provided to the bogie contoller (relevant for the ALS): – train speed – Drive direction – logical car number – car orientation – first car of the consist – 400 V AC is present – tbd in ICD document	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_03.Zefiro-Europe.CONCEPT.196	The following events related to ALS shall be triggerd depending on inputs from the bogie controller – status data – diagnostic data – configurationinfo of all ALS components (to be stored by TCMS, events tbd)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_03.Zefiro-Europe.CONCEPT.199	IP link from Bogie controller to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		

3.5 Diagnostics & CBM

3.5.1 Function Design

The CBM Controller monitors the conditions of certain bogie components to optimise the maintenance interventions by detecting degradations and failures.

3.5.2 Requirements

ID	2F_03.Zefiro-Europe.TRS.79
Description	The conditions of the bogie components shall be monitored continuously by a CBM system to optimise the behaviour in maintenance by early detecting degradations of e.g axle bearings, wheels and axles.
Safety level	N/A

ID	2F_03.Zefiro-Europe.TRS.60
Description	Periodical measurement analysis shall be done for the following components: bearings, wheels and axles. Notifications shall be given to the maintenance regime and if necessary also to the driver.
Safety level	SIL 0

CBM data is provided by the bogie monitoring equipment on the Ethernet to TCMS or directly to the Telediagnostica application.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_03.Zefiro-Europe.CONCEPT.104	The following data shall be provided to the Bogie Monitoring: – Trainspeed – train configuration, number of cars – GPS position	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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2F_03.Zefiro-Europe.CONCEPT.206	The following events shall be generated by TCMS: – tbd	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_03.Zefiro-Europe.CONCEPT.238	The following data from the bogie controller shall be handled: – real time process data to Telediganostica application – CBM data – tbd	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_03.Zefiro-Europe.CONCEPT.290	WFL: Enable wheel flange lubrication	L	2F_03.Zefiro-Europe.CONCEPT.157 2F_03.Zefiro-Europe.CONCEPT.350

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_03.Zefiro-Europe.CONCEPT.29	WFL: Spray impuls	L	2F_03.Zefiro-Europe.CONCEPT.167
2F_03.Zefiro-Europe.CONCEPT.291	WFL: System fault	L	2F_03.Zefiro-Europe.CONCEPT.159 2F_03.Zefiro-Europe.CONCEPT.351
2F_03.Zefiro-Europe.CONCEPT.297	IMS: 2 Digital outputs	L	2F_03.Zefiro-Europe.CONCEPT.129

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_03.Zefiro-Europe.CONCEPT.30	WFL: Spray impuls	L	2F_03.Zefiro-Europe.CONCEPT.280
2F_03.Zefiro-Europe.CONCEPT.352	WFL: System fault	L	2F_03.Zefiro-Europe.CONCEPT.351
2F_03.Zefiro-Europe.CONCEPT.294	HABD: Bogie supervision loop energized	L	2F_03.Zefiro-Europe.CONCEPT.257

HABD: Supervision of safety relais 1
HABD: Supervision of safety relais 2

Nr	Functional name	Type	Linked to requirement
2F_03.Zefiro-Europe.CONCEPT.295	HABD: Temperatur + ID of axle box	A	2F_03.Zefiro-Europe.CONCEPT.257
2F_03.Zefiro-Europe.CONCEPT.296	IMS: 2 Digital inputs	L	2F_03.Zefiro-Europe.CONCEPT.129 2F_03.Zefiro-Europe.CONCEPT.214
2F_03.Zefiro-Europe.CONCEPT.298	IMS: Supervision of IMS loop 2	L	2F_03.Zefiro-Europe.CONCEPT.215
2F_03.Zefiro-Europe.CONCEPT.337	IMS: Supervision of IMS loop 1	L	2F_03.Zefiro-Europe.CONCEPT.331
2F_03.Zefiro-Europe.CONCEPT.332	IMS: Instability present	L	2F_03.Zefiro-Europe.CONCEPT.214 2F_03.Zefiro-Europe.CONCEPT.322

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2F_03.Zefiro-Europe.CONCEPT.333	IMS: Instability loop at fault	L	2F_03.Zefiro-Europe.CONCEPT.321

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_03.Zefiro-Europe.CONCEPT.293	WFL: enable wheel flange lubrication system	L	2F_03.Zefiro-Europe.CONCEPT.108 2F_03.Zefiro-Europe.CONCEPT.350
2F_03.Zefiro-Europe.CONCEPT.292	WFL: activate solinoid valve	L	2F_03.Zefiro-Europe.CONCEPT.280

4.2 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_03.Zefiro-Europe.CONCEPT.32	ALS related data to be stored by TCMS – status data – diagnostic data – configurationinfo		2F_03.Zefiro-Europe.CONCEPT.196
2F_03.Zefiro-Europe.CONCEPT.300	CBM related data		2F_03.Zefiro-Europe.CONCEPT.104

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_03.Zefiro-Europe.CONCEPT.33	Train status data to be provided by TCMS – train speed – Drive direction – logical car number – car orientation – first car of the consist – 400 V AC is present – tbd in ICD document		2F_03.Zefiro-Europe.CONCEPT.195

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